

3(a)(i)	any three from: 1 PCR to amplify DNA ; 2 (gel) electrophoresis, to compare / separate, DNA fragments ; 3 microarray to, compare / analyse, genomes ; A polymorphisms 4 database / bioinformatics, to, compare / analyse, sequences of, (DNA) bases / nucleotides ;	3
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Question	Answer	Marks
3(a)(ii)	any four from: 1 <u>allopatric</u> (speciation) ; 2 <i>idea of no contact</i> so, reproductive isolation / no gene flow (between populations) ; 3 different mutations occur (in each population) ; 4 different selection pressures (for each population) ; 5 <i>ref. to</i> (named) morphological / (named) behavioural / <u>ecological</u> / <u>niche</u>, differences (between populations) ; 6 so can no longer interbreed to produce <u>fertile</u> offspring ;	4
3(a)(iii)	1 they are now different species / speciation has taken place ; 2 (different mating calls shows) <u>behavioural</u>, isolation / separation ; 3 (when S female and N male mate) offspring not produced / (fertilised) eggs are not viable ;	3
3(b)	1 <u>genes</u> / <u>genetic</u> and <u>environment</u>(al) ; 2 polygenes / many genes involved ; 3 diet / food (availability) / competition for food / nutrients / temperature / disease ;	3